

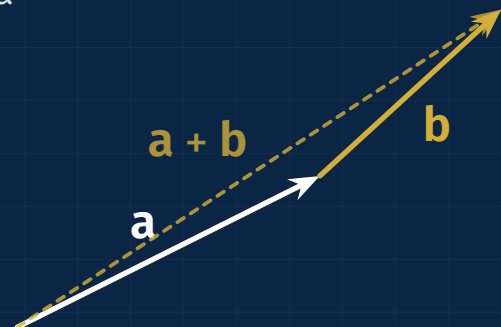


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CHAPTER 5 • VECTORS & TRANSFORMATIONS • TOPIC 1

Vectors

A vector is a journey. Write any route as a chain of known vectors; for a point on a line, go part of the way; where two lines meet, two routes must agree.



5

VARIANTS

28

PAST-PAPER PARTS

45

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Column vectors & magnitude

V2 Vector paths

V3 Ratios on a line

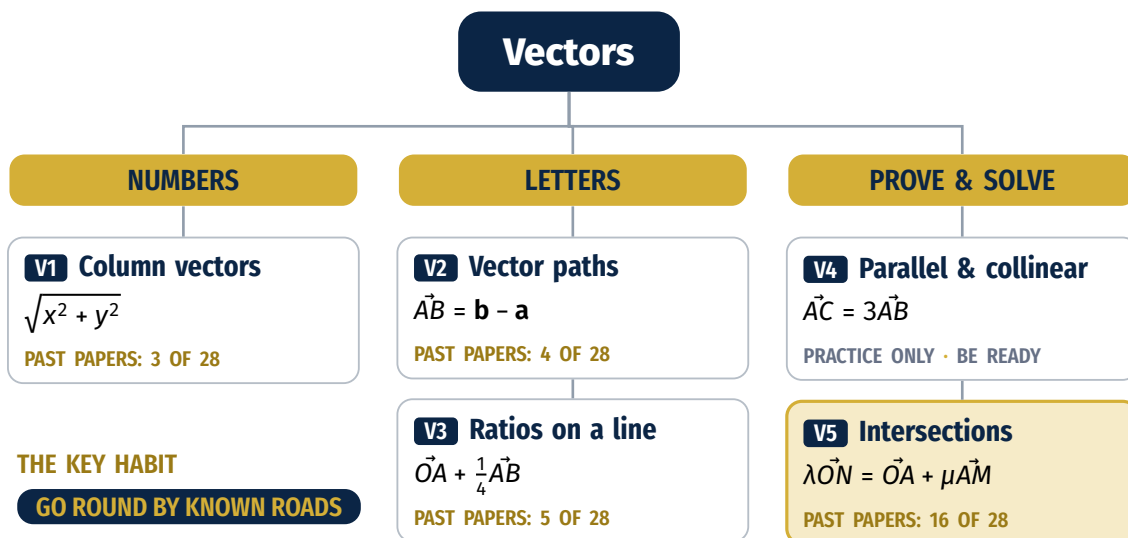
V4 Parallel & collinear proofs

V5 Intersections

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Vectors at a glance

 Your notes

**THE KEY HABIT****GO ROUND BY KNOWN ROADS**

Against an arrow? Use a minus sign.

★ How often does each variant come up?

20 past-paper questions (Jan 2020 → Nov 2025) hold 28 parts; 45 practice parts — grouped by type.

	PAST PAPERS (OUT OF 28)	PRACTICE PARTS
V1 Column vectors	3	15
V2 Vector paths	4	11
V3 Ratios on a line	5	4
V4 Parallel, collinear	0	15
V5 Intersections	16	0

WHAT THIS TELLS YOU On every paper, near the end (Q19–27, 5–6 marks): “using a vector method, find the ratio / find $\vec{OP}$ ”. Paths and ratios are the first marks; the two-routes equation finishes it.

BEFORE YOU START

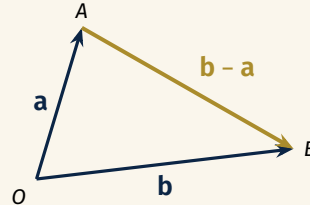
The big idea

A vector is a journey with a direction.

To get from A to B, go any way you like along vectors you know:

$$\vec{AB} = \vec{AO} + \vec{OB} = -\mathbf{a} + \mathbf{b}.$$

Going **against** an arrow: change the sign.



 Your notes



The rules

Reverse

$$\vec{BA} = -\vec{AB}$$

Change every sign.

Chain

$$\vec{AC} = \vec{AB} + \vec{BC}$$

End of one is start of the next.

Parallel

$$\mathbf{u} = k\mathbf{v}$$

One is a multiple of the other.

Magnitude

$$\left| \begin{pmatrix} x \\ y \end{pmatrix} \right| = \sqrt{x^2 + y^2}$$

Length (Pythagoras).



Point on a line

$$AP : PB = 1 : 3$$

$$\vec{OP} = \vec{OA} + \frac{1}{4}\vec{AB}$$

P is $\frac{1}{4}$ of the way from A.

Midpoint M of AB

$$\vec{OM} = \frac{1}{2}(\vec{OA} + \vec{OB})$$

Half way.

**CALCULATOR FIRST, THEN CHECK**

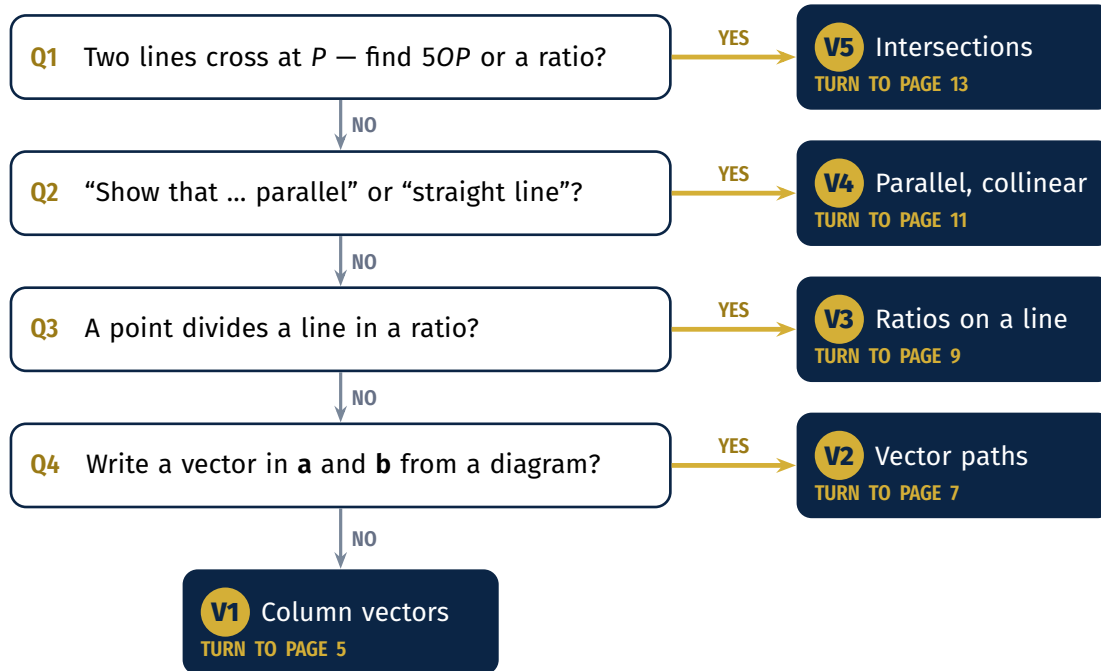
Collect **a** terms and **b** terms separately — like algebra with two letters.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Column Vectors & Magnitude

Add/subtract top with top, bottom with bottom; magnitude by Pythagoras.

SEEN IN PAST PAPERS

3 / 28

PRACTICE QUESTIONS: 15

 Your notes

Came up in: Jan 2021 P1HR Q13 • May 2023 P1HR Q19 • Nov 2024 P2H Q17

? What is it?

In plain words

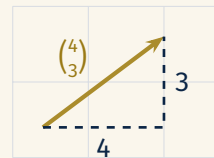
$\begin{pmatrix} x \\ y \end{pmatrix}$: x across, y up. Add or subtract **component by component**; multiply every component by a number.

Magnitude (length) = $\sqrt{x^2 + y^2}$.
Reversing a vector changes both signs.

IT LOOKS LIKE

$$\vec{AB} + \vec{BC}$$

$$|\vec{HF}|$$



$$\text{length } \sqrt{4^2 + 3^2} = 5$$

How do I do it?

1 Route

Write the vector as a chain: $\vec{AC} = \vec{AB} + \vec{BC}$.

2 Reverse

$\vec{BC} = -\vec{CB}$: flip both signs.

3 Add

Top with top, bottom with bottom.

4 Length

$$\sqrt{x^2 + y^2}.$$

TOP WITH TOP Then Pythagoras for the length.

V1 Column Vectors & Magnitude

VARIANT 1 • SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

“Find, as a column vector, $\vec{AC}$ ”→ $\vec{AB} + \vec{BC}$; reverse if needed

“Calculate the magnitude of the vector”

→ $\sqrt{x^2 + y^2}$

“how much further Pru travels”

→ Sum of lengths – direct length



WORKED EXAMPLE PAST PAPER

May 2023 • Paper 1HR • Q19 • 5 marks

$\vec{AB} = \begin{pmatrix} 7 \\ -2 \end{pmatrix}$, $\vec{BC} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$ (km). Pru goes $A \rightarrow B \rightarrow C$; Yang goes straight from A to C. How much further does Pru travel (1 d.p.)?

STEP 1

$$|\vec{AB}| = \sqrt{53}, \quad |\vec{BC}| = \sqrt{34}$$

◀ Pru's two legs

STEP 2

$$\vec{AC} = \begin{pmatrix} 7 \\ -2 \end{pmatrix} + \begin{pmatrix} -3 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

◀ Yang's route

STEP 3

$$|\vec{AC}| = \sqrt{16 + 9} = 5$$

◀ magnitude

STEP 4

$$\sqrt{53} + \sqrt{34} - 5 = 8.11$$

◀ difference

ANSWER 8.1 km

✓ direct route is shorter ✓



WATCH OUT

× $\vec{AC} = \vec{AB} + \vec{CB}$ ✓ $\vec{AB} + \vec{BC}$, and $\vec{BC} = -\vec{CB}$

The chain must join end to start.



EXAM TIP

Magnitudes don't add like vectors: add the lengths only when the journey has two legs.



Check yourself

calculator first • answers on page 15

1.1

Jan 2021 P1HR

$\vec{AB} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$ and $\vec{CB} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$.
Find $\vec{AC}$ as a column vector.

Answer:

1.2

Nov 2024 P2H

$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$ and $\vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$.
Calculate the magnitude of $\vec{HF}$.

Answer:

1.3

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$\mathbf{a} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$. Find $2\mathbf{a} + \mathbf{b}$ and its magnitude.

Answer:

V2

VARIANT 2

Vector Paths

Find a route made of vectors you know;
collect **a** and **b**.

SEEN IN PAST PAPERS

4 / 28

PRACTICE QUESTIONS: 11

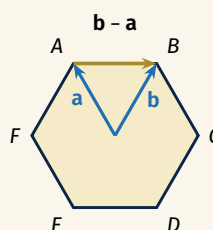

Your notes

Came up in: Nov 2021 P1H Q14 • Jun 2025 P2HR Q18

? What is it?**In plain words**

To write $\vec{XY}$ in **a** and **b**: start at X,
follow **known** arrows to Y. With an
arrow: +. Against: −.

Parallel sides of a shape are
multiples: an enlargement scale 2
doubles every vector.

IT LOOKS LIKE $\vec{AB}$ $\vec{KI}$ $\vec{LD}$ in a hexagon**How do I do it?****1 Start, end**

Where does the vector go from and to?

2 Route

Along known vectors only.

3 Signs

With an arrow +, against −.

4 SimplifyCollect **a**'s and **b**'s.**GO ROUND BY KNOWN ROADS** Any route gives the same answer.

V2 Vector Paths

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

“Write down $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ ” → $\vec{AO} + \vec{OB} = \mathbf{b} - \mathbf{a}$

“GHIJKL is an enlargement ... scale factor 2” → Each big vertex = $2 \times$ the small one

“Simplify your answers” → Collect like terms

 WORKED EXAMPLE **PAST PAPER**

June 2025 · Paper 2HR · Q18 · 3 marks

$\vec{OA} = 4\mathbf{a}$, $\vec{OB} = 4\mathbf{b}$. P is on AB with $AP : PB = 1 : 3$. (a) Write $\vec{AB}$. (b) Express $\vec{OP}$ in simplest form.

STEP 1 $\vec{AB} = -4\mathbf{a} + 4\mathbf{b} = 4\mathbf{b} - 4\mathbf{a}$ ◀ (a) $A \rightarrow O \rightarrow B$

STEP 2 $\vec{AP} = \frac{1}{4}\vec{AB} = \mathbf{b} - \mathbf{a}$ ◀ a quarter of the way

STEP 3 $\vec{OP} = \vec{OA} + \vec{AP} = 4\mathbf{a} + \mathbf{b} - \mathbf{a}$ ◀ route $O \rightarrow A \rightarrow P$

STEP 4 $= 3\mathbf{a} + \mathbf{b}$ ◀ simplify

ANSWER (a) $4\mathbf{b} - 4\mathbf{a}$ (b) $3\mathbf{a} + \mathbf{b}$

✓ P is near A : mostly $\mathbf{a}$ ✓

 WATCH OUT

✗ $\vec{AB} = 4\mathbf{a} + 4\mathbf{b}$

✓ $4\mathbf{b} - 4\mathbf{a}$

From A to O is against the arrow $\vec{OA}$.

 EXAM TIP

Write the route in letters first:

$\vec{OP} = \vec{OA} + \vec{AP}$.

 Check yourself

calculator first · answers on page 15

2.1 Nov 2021 P1H

Regular hexagon $ABCDEF$, centre O ,
 $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$; $GHIJKL$
 is its enlargement $\times 2$
 about O . Find $\vec{LD}$.

Answer:

2.2 Nov 2021 P1H

Same hexagons;
 triangle OAB has area 5
 cm^2 . Area between the
 two hexagons?

Answer:

2.3 Dr Eslam

$\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$. Find $\vec{BA}$.

Answer:

V3

VARIANT 3

Ratios on a Line

A point $\frac{m}{m+n}$ of the way along: start at one end, add that fraction of the line.

SEEN IN PAST PAPERS

5 / 28

PRACTICE QUESTIONS: 4

 Your notes

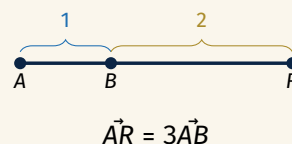
Came up in: Jun 2022 P2HR Q25 • Jun 2025 P1H Q23 • Nov 2025 P1H Q27

? What is it?

In plain words

$AB : BR = 1 : 2$ means R is beyond B with $AR = 3AB$. $BD : DC = 1 : 3$ means D is $\frac{1}{4}$ of the way from B .

With an **unknown** (n, λ) : write the point two ways and **compare** **a**'s and **b**'s.



IT LOOKS LIKE

$$AB : BR = 1 : 2$$

$$\vec{OQ} = n\mathbf{b}$$

How do I do it?

1 Fraction

Ratio $m : n \rightarrow \frac{m}{m+n}$ of the line.

2 Point

Start + fraction $\times$ line vector.

3 Unknown?

Write it a second way (it is on another line too).

4 Compare

a with **a**, **b** with **b**; solve.

START + FRACTION $\times$ LINE Then compare coefficients.

V3 Ratios on a Line

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO“ $AB : BR = 1 : 2 \dots \vec{OQ} = n\vec{b}$ ”→ **R from A; Q on PR and on OB**“ $\vec{OP} = n\vec{OD}$ ”→ **Two expressions for $\vec{OP}$**

“KLM is a straight line”

→ **$\vec{KL}$ parallel to $\vec{KM}$** **WORKED EXAMPLE PAST PAPER**

June 2025 • Paper 1H • Q23 • 5 marks

$\vec{OA} = 12\vec{a}$, $\vec{OB} = 8\vec{b}$; P is the midpoint of OA; ABR and PQR are straight with $AB : BR = 1 : 2$; Q is on OB with $\vec{OQ} = n\vec{b}$. Find n.

STEP 1

$$\vec{AB} = 8\vec{b} - 12\vec{a}, \vec{OR} = 12\vec{a} + 3\vec{AB} = 24\vec{b} - 24\vec{a} \quad \blacktriangleleft \text{(a) and R}$$

STEP 2

$$\vec{PR} = 24\vec{b} - 30\vec{a} \quad \blacktriangleleft \vec{OR} - \vec{OP}, \vec{OP} = 6\vec{a}$$

STEP 3

$$\vec{OQ} = 6\vec{a} + t(24\vec{b} - 30\vec{a}) \quad \blacktriangleleft \text{Q on PR}$$

STEP 4

$$\text{a: } 6 - 30t = 0 \Rightarrow t = \frac{1}{5}; n = \frac{24}{5} \quad \blacktriangleleft \text{Q on OB: no a}$$

ANSWER $n = \frac{24}{5}$

✓ $\vec{OQ} = 4.8\vec{b}$, inside $OB = 8\vec{b}$ ✓

**WATCH OUT**

✗ $\vec{AR} = 2\vec{AB}$

✓ $\vec{AR} = 3\vec{AB}$

$AB : BR = 1 : 2$ — the whole line AR is 3 parts.

**EXAM TIP**

“On OB” means “no a part” — that gives the equation.

**Check yourself**

calculator first • answers on page 15

3.1

Nov 2025 P1H

$\vec{OA} = 2\vec{a}$, $\vec{AB} = 4\vec{b} - \vec{a}$,
 $\vec{OC} = 5\vec{b} - 3\vec{a}$;
 $BD : DC = 1 : 3$; ABP,
 ODP straight; $\vec{OP} = n\vec{OD}$.
 Find n.

Answer:

3.2

Jun 2022 P2HR

Parallelogram ABCD,
 $\vec{AB} = \vec{a}$, $\vec{BC} = \vec{b}$, $\vec{DM} = \frac{1}{2}\vec{b}$;
 $AK : AB = \lambda : 1$,
 $CL : CD = \mu : 1$; KLM
 straight; $\lambda : \mu = 1 : 2$.

Answer:

3.3

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$\vec{OA} = 6\vec{a}$, $\vec{OB} = 6\vec{b}$,
 $AP : PB = 1 : 2$. Find
 $\vec{OP}$.

Answer:

V4

VARIANT 4

Parallel & Collinear Proofs

Parallel: one vector is a number times the other. Collinear: parallel and sharing a point.

SEEN IN PAST PAPERS

0 / 28

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 15


Your notes

? What is it?

In plain words

Two vectors are **parallel** if one is a **multiple** of the other:

$$6\mathbf{a} - 9\mathbf{b} = 3(2\mathbf{a} - 3\mathbf{b}).$$

Three points are on a **straight line** if $\vec{AC} = k\vec{AB}$ — parallel **and** through the same point A.

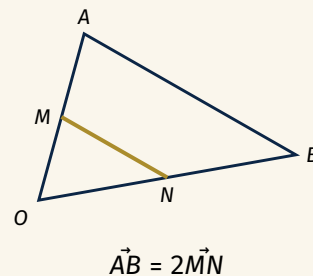
IT LOOKS LIKE

 $MN \parallel AB$

A

B

C collinear



How do I do it?

1 Both vectors

Write each in **a**, **b**.

2 Factor

Take out a common factor.

3 Multiple?

Same bracket → parallel.

4 Conclude

Parallel (and a shared point → straight line).

SAME BRACKET = PARALLEL Shared point → collinear.

V4 Parallel & Collinear Proofs

VARIANT 4 · SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

“Show that MN is parallel to AB ” → $\vec{AB} = k\vec{MN}$

“Prove that A, B, C lie on a straight line” → $\vec{AC} = k\vec{AB}$ and the common point A

“Are they parallel?” → Not a multiple → No



WORKED EXAMPLE DR ESAM

Proof · 3 marks

$\vec{OA} = 2\mathbf{a}$, $\vec{OB} = 2\mathbf{b}$. M and N are the midpoints of OA and OB . Show that MN is parallel to AB and half its length.

STEP 1 $\vec{MN} = -\mathbf{a} + \mathbf{b} = \mathbf{b} - \mathbf{a}$ ◀ $M \rightarrow O \rightarrow N$

STEP 2 $\vec{AB} = 2\mathbf{b} - 2\mathbf{a}$ ◀ $A \rightarrow O \rightarrow B$

STEP 3 $= 2(\mathbf{b} - \mathbf{a}) = 2\vec{MN}$ ◀ factor

STEP 4 A multiple of $\vec{MN}$ ◀ parallel, twice as long

ANSWER $\vec{AB} = 2\vec{MN}$: parallel, $MN = \frac{1}{2}AB$ ✓ same bracket $(\mathbf{b} - \mathbf{a})$ ✓



WATCH OUT

× “they look parallel”

✓ show $\vec{AB} = 2\vec{MN}$

A vector proof needs the multiple written out.



EXAM TIP

For a straight line also say “and they share the point A ”.



Check yourself

calculator first · answers on page 15

4.1

Practice · Medium

$\vec{OM} = \mathbf{m}$, $\vec{ON} = \mathbf{n}$,
 $\vec{OA} = 2\mathbf{m}$, $\vec{OB} = 2\mathbf{n}$. Write
 $\vec{AB}$ in terms of $\vec{MN}$.

Answer:

4.2

Practice · Hard

$\vec{BC} = 3\mathbf{b} - 2\mathbf{a}$ and
 $\vec{CD} = -10\mathbf{b}$. Are BC and
 CD parallel?

Answer:

4.3

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Show that $6\mathbf{a} - 9\mathbf{b}$ is
parallel to $2\mathbf{a} - 3\mathbf{b}$.

Answer:

V5**VARIANT 5** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

16 / 28

PRACTICE QUESTIONS: 0

Intersections: Two Routes to One Point

P is on two lines: write $\vec{OP}$ along each with an unknown, then compare a and b .

Came up in: Jan 2020 P2HR Q26 • Jan 2020 P2H Q23 • Jan 2021 P1H Q23 • May 2021 P1H Q26 • Jan 2022 P1HR Q24 • Jan 2022 P1H Q22 • Jun 2022 P2H Q21 • May 2023 P1H Q19 • Nov 2023 P2H Q22 • Jun 2024 P2HR Q25 • Jun 2024 P2H Q24 • Nov 2024 P1H Q24

 Your notes

? What is it?

In plain words

P lies on **two lines**. Along line 1:

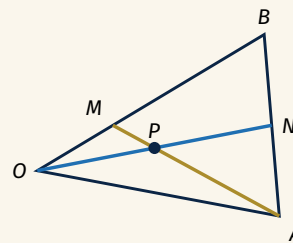
$\vec{OP} = \lambda \vec{ON}$. Along line 2:

$\vec{OP} = \vec{OA} + \mu \vec{AM}$.

Same point, so the

a parts are equal and the

b parts are equal: two equations, two unknowns.



IT LOOKS LIKE

find $\vec{OP}$ find $AP : PM$

How do I do it?

1 VectorsFind the two line vectors ($\vec{ON}$, $\vec{AM}$).**2** Route 1 $\vec{OP} = \lambda \vec{ON}$.**3** Route 2 $\vec{OP} = \vec{OA} + \mu \vec{AM}$.**4** Compare**a:** ..., **b:** ...; solve; substitute.**TWO ROUTES, ONE POINT** 16 of the 28 past-paper parts.

V5 Intersections: Two Routes to One Point

VARIANT 5 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"P is the point of intersection of ON and AM" → **Two routes to P**

"Using a vector method, find the ratio AP : PM" → **$\mu : (1 - \mu)$ from the route**

"OY = k × ON ... find k" → **k is the multiplier on one route**

WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 1H • Q22 • 5 marks

$\vec{OA} = 8\mathbf{a}$, $\vec{OB} = 6\mathbf{b}$; $OM : MB = 1 : 2$; N is the midpoint of AB; P is where ON meets AM. Find $\vec{OP}$.

STEP 1 $\vec{OM} = 2\mathbf{b}$, $\vec{ON} = 4\mathbf{a} + 3\mathbf{b}$, $\vec{AM} = 2\mathbf{b} - 8\mathbf{a}$ ◀ line vectors

STEP 2 $\lambda(4\mathbf{a} + 3\mathbf{b}) = 8\mathbf{a} + \mu(2\mathbf{b} - 8\mathbf{a})$ ◀ two routes to P

STEP 3 **a:** $4\lambda = 8 - 8\mu$; **b:** $3\lambda = 2\mu$ ◀ compare

STEP 4 $\mu = \frac{3}{2}\lambda \Rightarrow 16\lambda = 8$, $\lambda = \frac{1}{2}$ ◀ solve

ANSWER $\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$

✓ $\mu = \frac{3}{4}$: $8\mathbf{a} + \frac{3}{4}(2\mathbf{b} - 8\mathbf{a})$ gives the same ✓

WATCH OUT

✗ using the same letter on both routes

✓ λ on one, μ on the other

The fractions along the two lines are different.

EXAM TIP

"Show your working": write both **a** and **b** equations.

Check yourself

calculator first • answers on page 15

5.1 Jun 2022 P2H

$\vec{OA} = 4\mathbf{a}$, $\vec{OB} = 6\mathbf{b}$,
 $\vec{AP} = 2\mathbf{a} + 8\mathbf{b}$; Q on AB
 with OQP straight. Find
 AQ : QB.

Answer:

5.2 May 2021 P1H

Trapezium OACB:
 $\vec{OA} = 2\mathbf{a}$, $\vec{OB} = 5\mathbf{b}$,
 $\vec{AC} = 3\mathbf{b}$. Diagonals OC,
 AB meet at P. Find $\vec{OP}$.

Answer:

5.3 Jan 2020 P2H

$\vec{AB} = 4\mathbf{a}$, $\vec{AC} = 2\mathbf{b}$; M
 midpoint of BC;
 $\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$. Find
 AP : PM.

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 $\begin{pmatrix} 7 \\ -1 \end{pmatrix}$

1.2 15

1.3 $\begin{pmatrix} 7 \\ -2 \end{pmatrix}$; 7.28

2.1 $2\mathbf{b} - 3\mathbf{a}$

2.2 90 cm^2

2.3 $\mathbf{a} - \mathbf{b}$

3.1 $n = \frac{32}{17}$

3.2 $\lambda = \frac{3}{7}, \mu = \frac{6}{7}$

3.3 $4\mathbf{a} + 2\mathbf{b}$

4.1 $\vec{AB} = 2\vec{MN}$

4.2 No

4.3 $6\mathbf{a} - 9\mathbf{b} = 3(2\mathbf{a} - 3\mathbf{b})$

5.1 8 : 9

5.2 $\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$

5.3 3 : 1

THE WHOLE TOPIC ON ONE PAGE

Vectors cheat sheet



Your notes
V1 Column 3/28

$$\sqrt{x^2 + y^2}$$

Top with top; reverse = flip signs.

V2 Paths 4/28

$$\vec{AB} = \mathbf{b} - \mathbf{a}$$

Known roads; against an arrow –.

V3 Ratios 5/28

$$\vec{OA} + \frac{m}{m+n}\vec{AB}$$

Start + fraction of the line.

V4 Parallel

$$\vec{AC} = 3\vec{AB}$$

Multiple; shared point → straight.

V5 Intersections 16/28

$$\lambda\vec{ON} = \vec{OA} + \mu\vec{AM}$$

Compare **a** and **b**.**TOP 3 LOST MARKS**

1. sign wrong against an arrow × 2. $\frac{m}{n}$ instead of $\frac{m}{m+n}$ × 3. one letter for both routes ×

BOTTOM LINE Known roads · against an arrow, minus · same bracket = parallel · two routes, compare.